

Django-12_Module-2_Assignment

Objective

Build a command-line **Expense Tracker** application using Python.

Project Description

Create an Expense Tracker that allows users to record daily expenses, view expenses, search, update, delete, and generate a spending summary.

Store all expense data in a text file.

Expense Information

Each expense should contain:

- Expense ID
- Date (YYYY-MM-DD)
- Category
- Description
- Amount

Example:

1,2026-07-11,Food,Lunch at restaurant,350

2,2026-07-11,Transport,Rickshaw,80

3,2026-07-12,Shopping,T-Shirt,1200

Store all data inside:

expenses.txt

Menu

===== Expense Tracker =====

1. Add Expense
2. View All Expenses
3. Search Expense
4. Update Expense
5. Delete Expense
6. Expense Summary
7. Exit

Choose an option:

Functional Requirements

1. Add Expense

Ask the user for:

Expense ID:

Date (YYYY-MM-DD):

Category:

Description:

Amount:

Save the expense in `expenses.txt`.

Create the function:

```
def add_expense():
```

2. View All Expenses

Display all expenses in a readable format.

Example:

Expense ID : 1

Date : 2026-07-11

Category : Food

Description: Lunch

Amount : 350

Function:

```
def view_expenses():
```

3. Search Expense

Search by **Expense ID**.

If found:

Expense Found

Expense ID : 2

Date : 2026-07-11

Category : Transport

Description: Rickshaw

Amount : 80

Otherwise:

Expense not found.

Function:

```
def search_expense():
```

4. Update Expense

Search using Expense ID.

Allow the user to update:

- Date
- Category
- Description
- Amount

Function:

```
def update_expense():
```

5. Delete Expense

Delete an expense using Expense ID.

Function:

```
def delete_expense():
```

6. Expense Summary

Display:

- Total number of expenses
- Total amount spent
- Average expense
- Highest expense
- Lowest expense

Example:

===== Expense Summary =====

Total Expenses : 8

Total Spending : 7,850 BDT

Average Expense : 981.25 BDT

Highest Expense : 2,500 BDT

Lowest Expense : 80 BDT

Function:

```
def expense_summary():
```

7. Exit

Exit the program safely.

Exception Handling

Your program must handle the following exceptions.

Invalid Amount

If the user enters:

abc

Output:

Invalid amount! Please enter a valid number.

Negative Amount

Amount cannot be negative.

File Not Found

No expense records found.

Empty File

No expenses available.

Invalid Menu Option

Invalid choice! Please select a valid option.

Required Functions

`add_expense()`

`view_expenses()`

search_expense()

update_expense()

delete_expense()

expense_summary()

main()

Sample Output

===== Expense Tracker =====

1. Add Expense
2. View All Expenses
3. Search Expense
4. Update Expense
5. Delete Expense
6. Expense Summary
7. Exit

Choose: 1

Expense ID: 1

Date: 2026-07-11

Category: Food

Description: Lunch

Amount: 350

Expense added successfully.

Choose: 2

Expense ID : 1

Date : 2026-07-11

Category : Food

Description: Lunch

Amount : 350

Choose: 6

===== Expense Summary =====

Total Expenses : 3

Total Spending : 1630 BDT

Average Expense : 543.33 BDT

Highest Expense : 1200 BDT

Lowest Expense : 80 BDT

Submission Guidelines

- Just submit your python file.